Cambridge International AS & A Level

BIOLOGY (9700) P1

TOPIC WISE QUESTIONS + ANSWERS | COMPLETE SYLLABUS



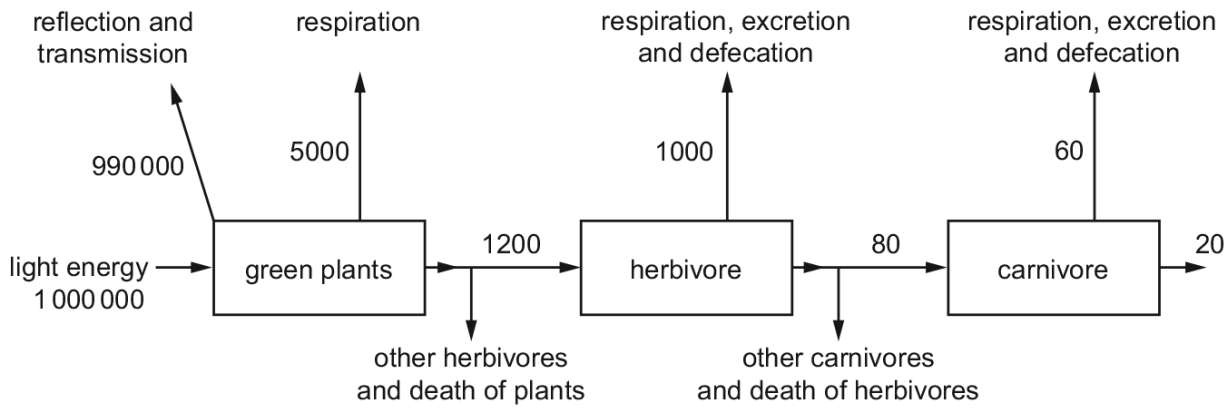
Chapter 13

Photosynthesis

13.1 Photosynthesis as an energy transfer process

1606. 9700_w15_qp_11 Q: 39

The diagram shows the flow of energy through a food chain. Figures are in kJ m^{-2} $year^{-1}$.

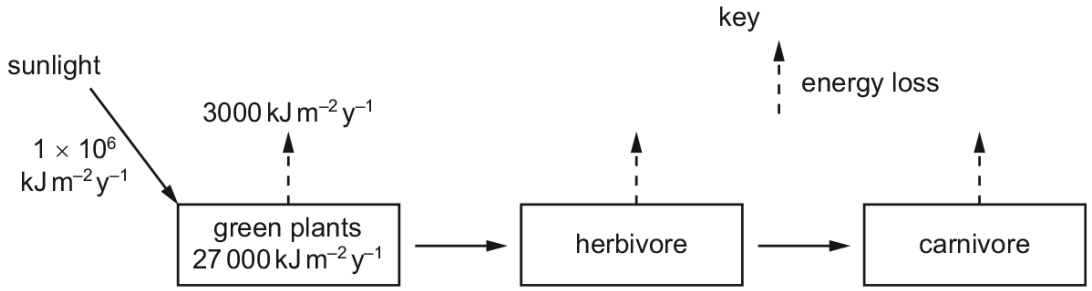


What percentage of light energy is available to the herbivore?

A 0.12 % **B** 0.38 % **C** 0.50 % **D** 1.00 %

1607. 9700_w15_qp_12 Q: 39

The diagram shows the flow of energy through a food chain.



What percentage of the energy is used by the green plants for photosynthesis?

A 0.03% **B** 0.30% **C** 2.70% **D** 3.00%

1608. 9700_w15_qp_13 Q: 39

The table shows the results of a field study of four species in a food chain in an area of woodland.

species	number of individuals	biomass of one individual / arbitrary units	energy value per unit mass / arbitrary units
R	10 000	0.1	1.0
S	5	10.0	2.0
T	500	0.002	1.8
U	3	300 000.0	0.5

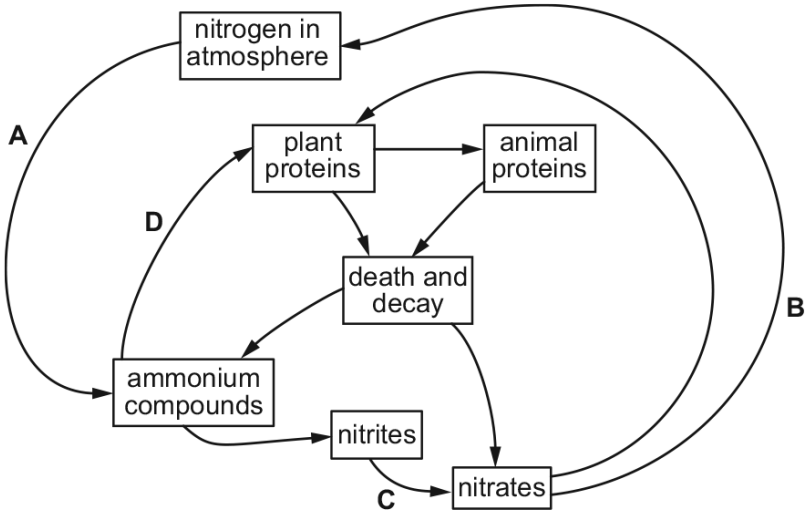
What is the energy flow in the food chain?

	from ⟶ to			
A	R	T	S	U
B	S	T	R	U
C	U	R	S	T
D	U	S	T	R

1609. 9700_w15_qp_13 Q: 40

The diagram shows a simplified nitrogen cycle.

Which arrow represents the activity of nitrogen-fixing bacteria?



13.2 Adaptations for photosynthesis

1610. 9700_s15_qp_12 Q: 38

At which stage of a food chain in a tropical grassland is the energy transfer least efficient?

